

Semantic Exploration Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Analysis of Lag-Based Semantic Distance

PENPAL Project

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1 Overview

The semantic exploration outputs now use the **current lag-based structure**:

- `k`: lag
- `distance`: mean semantic distance for that lag
- `agent`: "interleaved" for the full exchange sequence, or `author_1` / `author_2` for speaker-specific trajectories

The processed filename is still `semantic_exploration_binned.parquet`, but the current generator computes **lag-based distances**, not the older non-overlapping window-centroid bins.

This notebook therefore no longer assumes old turn-level exploration tables. Instead it analyses:

1. **Interleaved sequence exploration** across conditions
2. **Speaker-specific exploration** using recovered role metadata

Important comparison note:

- **Slot 1 / Slot 2** is the directly comparable cross-condition axis.
- **Human / AI** in Human-AI and **Starter-side / Non-starter-side** in same-type conditions are only used as within-condition descriptive views.

Primary cross-condition object. Per story, we compute the trapezoidal area under each slot’s mean distance(k) curve over the intersection of k values present for both slots, and take the unsigned difference:

$$A_{\text{semexp}}(i) = | \text{AUC}_{\text{slot}=1, i} - \text{AUC}_{\text{slot}=2, i} |.$$

This is label-invariant and comparable across conditions. The lag-binned LMM slot contrast and the log-distance slope view are retained as supplemental. Starter-mechanism and Human-AI speaker-type decompositions are deferred.

Article-use note. The primary claim should come from the per-story AUC asymmetry test. The interleaved-distance model answers a different question: whether the whole story sequence explores semantic space differently by condition, not whether the two contributors are more asymmetric.

2 Load Data

Table 1: Loaded semantic exploration rows by condition and agent view

condition	agent	n_stories	n_rows	min_k	max_k
Human-AI	author_1	97	577	2	7
Human-AI	author_2	97	577	2	7
Human-AI	interleaved	97	873	2	10
Human-Human	author_1	36	252	2	8
Human-Human	author_2	36	252	2	8
Human-Human	interleaved	36	324	2	10
AI-AI	author_1	80	560	2	8
AI-AI	author_2	80	560	2	8
AI-AI	interleaved	80	720	2	10

3 Preprocessing

Interleaved rows: 1917

Speaker rows: 2778

4 Primary: Per-Story AUC Asymmetry

Table 2: Stories with per-story AUC computable in both slots

condition	n_stories
Human-AI	97
Human-Human	36
AI-AI	80

Table 3: Per-story $|AUC(\text{slot } 1) - AUC(\text{slot } 2)|$ by condition

condition	n	mean_A	sd_A	median_A
Human-AI	97	0.191	0.128	0.175
Human-Human	36	0.172	0.136	0.157
AI-AI	80	0.180	0.147	0.165

```
## One-way ANOVA: A_semexp ~ condition
##           Df Sum Sq Mean Sq F value Pr(>F)
## condition    2  0.011  0.005488   0.294  0.745
## Residuals  210  3.914  0.018639
##
## Kruskal-Wallis (robust): A_semexp ~ condition
##
## Kruskal-Wallis rank sum test
##
## data: A_semexp by condition
## Kruskal-Wallis chi-squared = 1.3757, df = 2, p-value = 0.5026
##
##
## Pairwise Wilcoxon (BH-adjusted):
##
## Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data: df_auc_stories$A_semexp and df_auc_stories$condition
##
##           Human-AI Human-Human
## Human-Human 0.54      -
## AI-AI       0.54      0.97
##
## P value adjustment method: BH
```

Per-story semantic-exploration AUC asymmetry by condition

$$A_semexp(i) = |AUC_{\{slot\ 1\}}(i) - AUC_{\{slot\ 2\}}(i)|$$

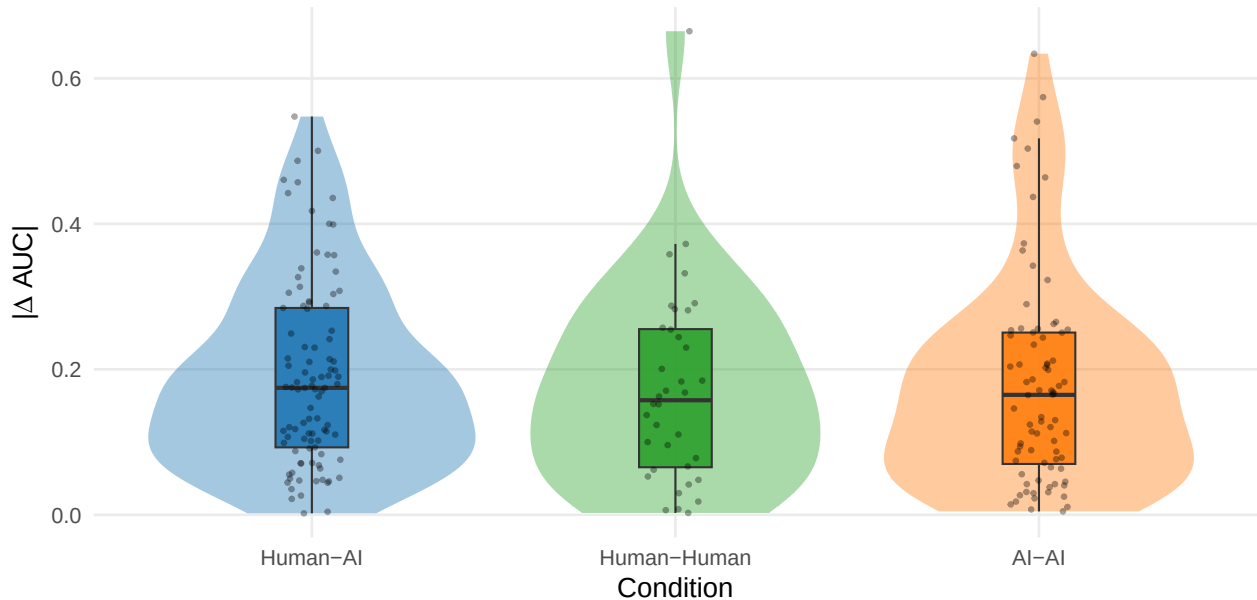


Table 4: Secondary: per-story $|\log(\text{distance}) \sim \log(k) \text{ slope diff}|$ by condition

condition	n	mean_A	median_A
Human-AI	97	0.148	0.113
Human-Human	36	0.122	0.095
AI-AI	80	0.110	0.080

5 Bootstrap CIs and Effect Sizes on A_semexp

Table 5: Bootstrap 95% CI on per-condition mean of A_semexp (R = 10,000)

condition	n	mean	ci_lower	ci_upper
Human-AI	97	0.191	0.166	0.217
Human-Human	36	0.172	0.131	0.218
AI-AI	80	0.180	0.150	0.214

Table 6: Pairwise Cliff’s delta on A_semexp

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	97	36	0.104	negligible
Human-AI vs AI-AI	97	80	0.088	negligible
Human-Human vs AI-AI	36	80	-0.005	negligible

6 Planned Contrast: HA vs (HH + AA)/2

Table 7: Planned contrast: A_semexp in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	0.015	0.019	0.779	161.318	0.437	-0.023	0.053

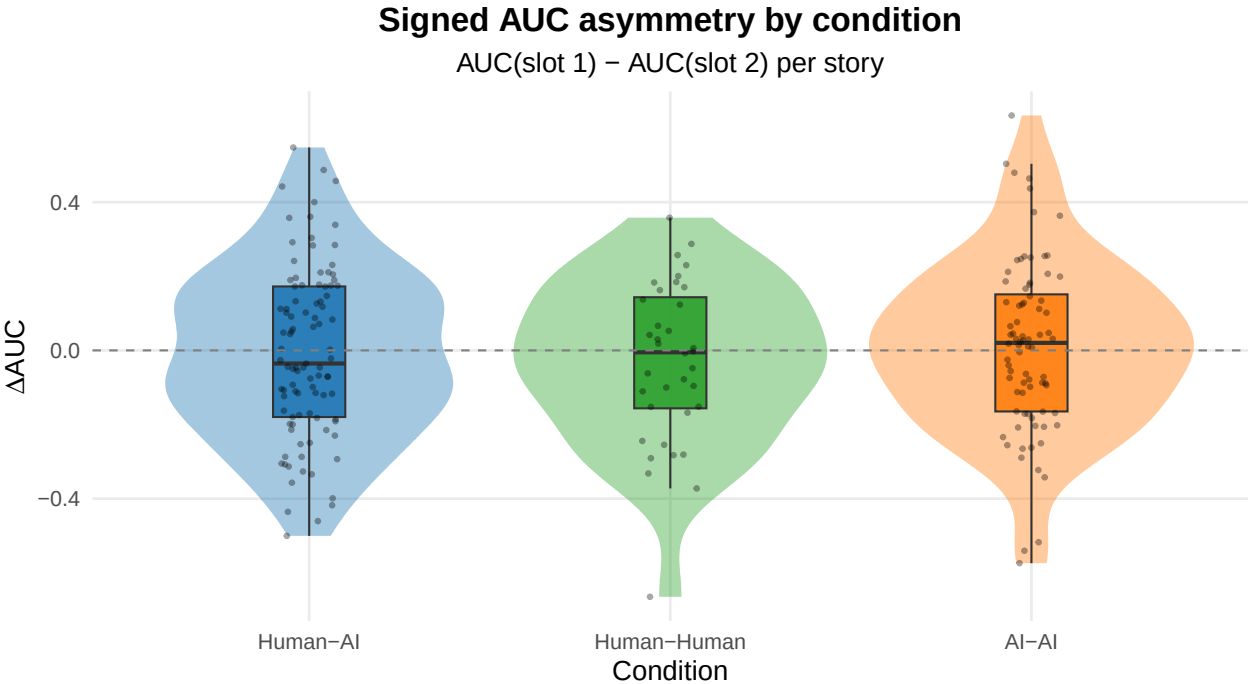
7 Directional Stability of Signed Semantic-Exploration AUC Asymmetry

Tests whether the *direction* of the slot-1 vs slot-2 AUC difference is consistent across stories within each condition. The signed quantity is $\text{signed}_i = \text{AUC}_{\text{slot } 1, i} - \text{AUC}_{\text{slot } 2, i}$.

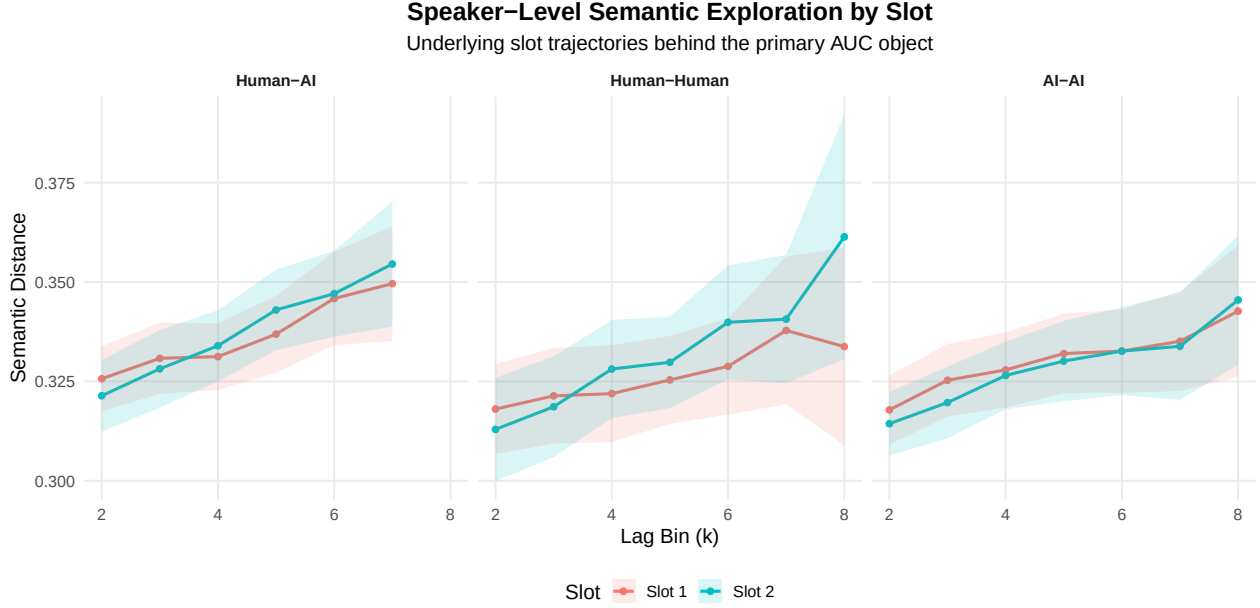
Table 8: Per-condition directional stability of signed AUC asymmetry

condition	n	mean	sd	stability	sign_consistency	one_sample_t_p
Human-AI	97	-0.007	0.231	0.028	0.526	0.782
Human-Human	36	-0.033	0.219	0.151	0.528	0.371
AI-AI	80	0.011	0.233	0.045	0.550	0.687

Brown-Forsythe variance-equality p = 0.7736



8 Supplemental: Speaker-Level Slot Trajectories

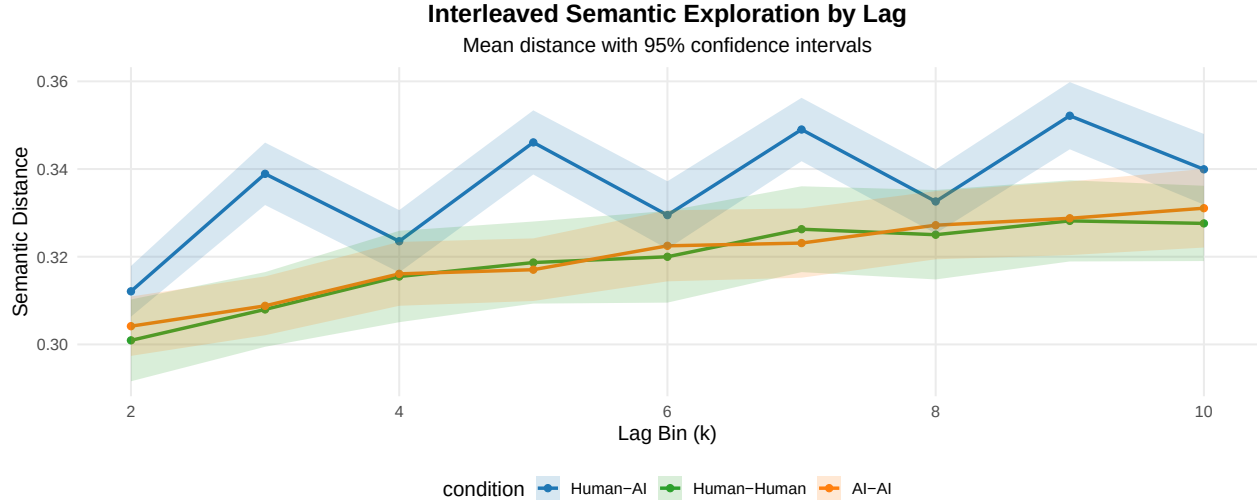


9 Supplemental: General Condition-Level Comparison

Table 9: Interleaved semantic distance by lag and condition

condition	k	mean_distance	sd_distance	n
Human-AI	2	0.312	0.029	97
Human-AI	3	0.339	0.036	97
Human-AI	4	0.324	0.036	97
Human-AI	5	0.346	0.037	97
Human-AI	6	0.329	0.039	97
Human-AI	7	0.349	0.036	97
Human-AI	8	0.333	0.037	97
Human-AI	9	0.352	0.038	97
Human-AI	10	0.340	0.040	97
Human-Human	2	0.301	0.029	36
Human-Human	3	0.308	0.026	36
Human-Human	4	0.315	0.032	36
Human-Human	5	0.319	0.029	36
Human-Human	6	0.320	0.032	36
Human-Human	7	0.326	0.030	36
Human-Human	8	0.325	0.031	36
Human-Human	9	0.328	0.028	36
Human-Human	10	0.328	0.026	36
AI-AI	2	0.304	0.031	80
AI-AI	3	0.309	0.031	80
AI-AI	4	0.316	0.033	80
AI-AI	5	0.317	0.033	80
AI-AI	6	0.322	0.037	80
AI-AI	7	0.323	0.036	80
AI-AI	8	0.327	0.035	80

condition	k	mean_distance	sd_distance	n
AI-AI	9	0.329	0.038	80
AI-AI	10	0.331	0.041	80



```
## Interleaved model summary:
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: distance ~ k * condition + (1 | conversation_id)
## Data: df_interleaved
##
## REML criterion at convergence: -9389.1
##
## Scaled residuals:
##   Min       1Q   Median       3Q      Max
## -3.7835 -0.5532  0.0144  0.5725  3.3368
##
## Random effects:
## Groups          Name          Variance Std.Dev.
## conversation_id (Intercept) 0.0009747 0.03122
## Residual              0.0002878 0.01697
## Number of obs: 1917, groups: conversation_id, 213
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)   3.188e-01  3.487e-03 2.872e+02  91.415 < 2e-16 ***
## k              2.869e-03  2.224e-04 1.701e+03 12.902 < 2e-16 ***
## conditionHuman-Human -1.926e-02  6.702e-03 2.872e+02 -2.873 0.004366 **
## conditionAI-AI      -1.848e-02  5.187e-03 2.872e+02 -3.564 0.000428 ***
## k:conditionHuman-Human 3.634e-04  4.275e-04 1.701e+03  0.850 0.395325
## k:conditionAI-AI      3.938e-04  3.308e-04 1.701e+03  1.191 0.233990
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) k      cndH-H cAI-AI k:cH-H
## k              -0.383
```

```

## cndtnHmn-Hm -0.520  0.199
## condtnAI-AI -0.672  0.257  0.350
## k:cndtnHm-H  0.199 -0.520 -0.383 -0.134
## k:cndtAI-AI  0.257 -0.672 -0.134 -0.383  0.350

##
## Condition contrasts (interleaved view):
## contrast estimate SE df t.ratio p.value
## (Human-AI) - (Human-Human) 0.017077 0.00619 210 2.758 0.0190
## (Human-AI) - (AI-AI) 0.016122 0.00479 210 3.364 0.0027
## (Human-Human) - (AI-AI) -0.000955 0.00637 210 -0.150 1.0000
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests

```

10 Supplemental: Role-Level View

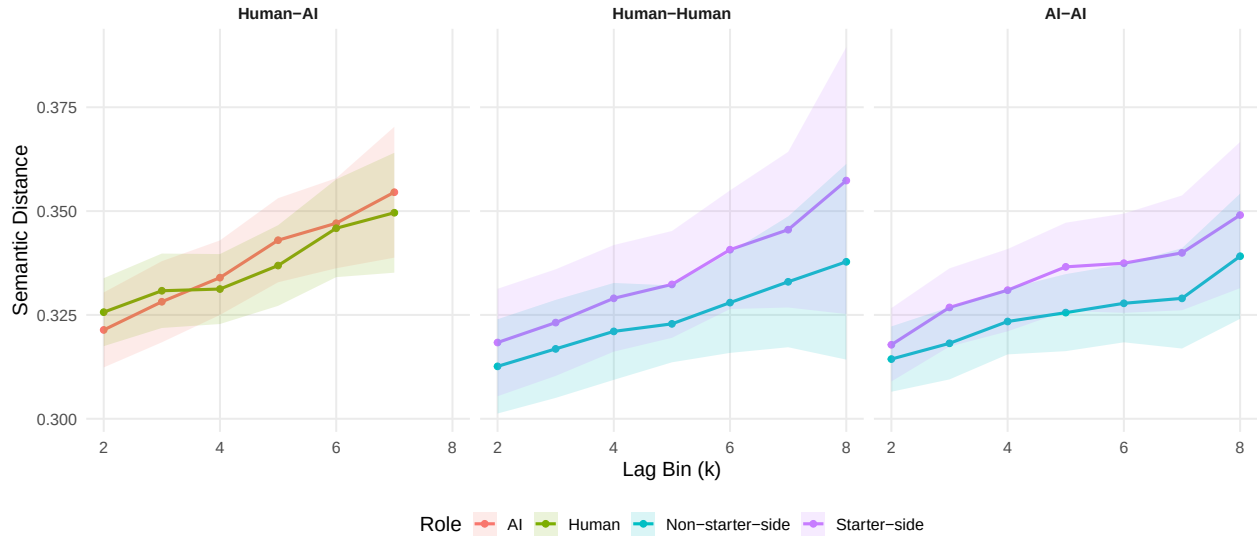
Table 10: Speaker-level semantic distance by condition, role, and lag

condition	condition_role	k	mean_distance	n
Human-AI	AI	2	0.321	97
Human-AI	AI	3	0.328	97
Human-AI	AI	4	0.334	97
Human-AI	AI	5	0.343	97
Human-AI	AI	6	0.347	96
Human-AI	AI	7	0.355	93
Human-AI	Human	2	0.326	97
Human-AI	Human	3	0.331	97
Human-AI	Human	4	0.331	97
Human-AI	Human	5	0.337	97
Human-AI	Human	6	0.346	96
Human-AI	Human	7	0.350	93
Human-Human	Non-starter-side	2	0.313	36
Human-Human	Non-starter-side	3	0.317	36
Human-Human	Non-starter-side	4	0.321	36
Human-Human	Non-starter-side	5	0.323	36
Human-Human	Non-starter-side	6	0.328	36
Human-Human	Non-starter-side	7	0.333	36
Human-Human	Non-starter-side	8	0.338	36
Human-Human	Starter-side	2	0.318	36
Human-Human	Starter-side	3	0.323	36
Human-Human	Starter-side	4	0.329	36
Human-Human	Starter-side	5	0.332	36
Human-Human	Starter-side	6	0.341	36
Human-Human	Starter-side	7	0.346	36
Human-Human	Starter-side	8	0.357	36
AI-AI	Non-starter-side	2	0.314	80
AI-AI	Non-starter-side	3	0.318	80
AI-AI	Non-starter-side	4	0.323	80
AI-AI	Non-starter-side	5	0.326	80
AI-AI	Non-starter-side	6	0.328	80

condition	condition_role	k	mean_distance	n
AI-AI	Non-starter-side	7	0.329	80
AI-AI	Non-starter-side	8	0.339	80
AI-AI	Starter-side	2	0.318	80
AI-AI	Starter-side	3	0.327	80
AI-AI	Starter-side	4	0.331	80
AI-AI	Starter-side	5	0.337	80
AI-AI	Starter-side	6	0.337	80
AI-AI	Starter-side	7	0.340	80
AI-AI	Starter-side	8	0.349	80

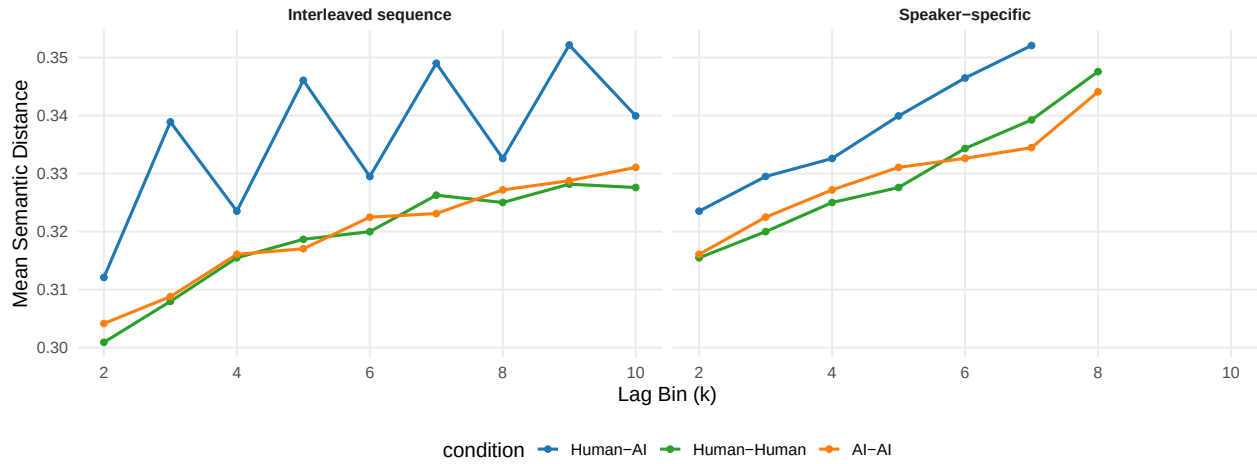
Speaker-Level Semantic Exploration by Role

Within-condition role view only: Human / AI in Human-AI; Starter-side / Non-starter-side in same-type conditions



11 Supplemental: Interleaved vs Speaker View

Interleaved vs Speaker-Specific Semantic Distance



12 Summary

Primary cross-condition result: per-story $A_{\text{semexp}} = |\text{AUC}_{\text{slot 1}} - \text{AUC}_{\text{slot 2}}|$ from trapezoidal integration of the distance(k) curve over the intersected k range. Tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon. A secondary log-distance-slope asymmetry is reported for dynamic-shape continuity with the paper.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of A_{semexp}
- pairwise Cliff's δ effect sizes
- planned single-df contrast HA vs (HH+AA)/2

Supplemental views retained:

- speaker-level slot trajectories (descriptive, both slots side-by-side without signed contrast)
- interleaved sequence model and condition descriptives
- within-condition role curves (descriptive only; HA Human/AI inferential view lives in analysis/human-ai/exploration.Rmd)
- interleaved vs speaker-specific side-by-side view

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized AUC asymmetry across all conditions)

Analysis completed: 2026-05-06 16:09:44.340388